Rajasthan **Board Class 12**

Subject – Chemistry

Time: $3\frac{1}{4}\,Hours$ **M.M: 56**

1. **Write any one example of network solid.**
   Sol:
   Graphite
2. **Write definition of aziotropic mixture.**
   Sol:
   Aziotropic is termed as the mixture of two liquid whose chemical composition is same both in liquid phase and vapour phase and whose boiling point is same nad boils at same temperature.
3. **Rate constant of chemical reaction is** $1.72\times10^{-4}g^{-1}$. **Calculate the order of reaction.**
   Sol:
   $$k = 1.72\times10^{-4}S^{-1}$$
   $$oder = Firstoder$$
4. **Define threshold energy.**
   Sol:
   The minimum amount of energy the must be possess by reactant molecule so that they can convert into product by collide with each other as known as threshold energy.

5. **Give any one example of bidentate ligand.**
   Sol:
   $C_2O_4^{-2}\left(oxolate\right)$
   $$\begin{array}{c} COO^- \\ | \\ COO^- \end{array}$$
6. **Write IUPAC name of Diethyl ether.**
   Sol:
   Ethoxy Ethane

7. **Draw the resonating structures of phenoxide ion.**
   Sol: Figure

8. **Write chemical equation of carbylamines reaction.**
   Sol:

$$R-NH_2+CHCl_3+3KOH$$
$$\downarrow$$
$$R-NC+3KCl+3H_2O$$

9. Write the name of the harmone secreted by thyroid gland.

Sol:

Thyroxin

10. Give any one example bio degradable polymer.

Sol:

Nylone-2 Nylone-6,

11. Write the monomer Buna-N polymer.

Sol.

(i) $CH_2=CH-CH=CH_2\left(Beta-1,3diene\right)$

(ii) $\underset{\displaystyle CN}{\underset{|}{CH_2=CH}}$ (Acrylonitrile)

12. What is polydispersity index.

Sol.

The ratio of weight average molecular mass and the number average molecular mass of polymer as known as polydispersity index.

$$P.D.I=\frac{\overline{M}_w}{\overline{M}_N}$$

13. Write the value of consent axis which presence in H_2O.

Sol.

The value of consent axis presence of H_2O in $C_x=2$

Section B

14. Write any two difference between shortkey franker defect.

Sol. Due to shortkey defect density of crystal is decrease why due to franker defect density of crystal remain unchanged there is no effect on the density of crystal franker effect.

Shortkey is produced is anion crystal due to the absence of equal number catoion anion while franker defect is kind of dislocation defect which is produced is anion crystal due to the change of position of catoion from it proper position to enter crystal sit.

Or,

15. $K_4[Fe(CN)_6]$, i=0.05m, α=92%=0.92 **then** $0.082 later\ mol^{-1}k^{-1}$ π=?

Sol.

We know that,

$$\pi = iCRT$$
$$\alpha = \frac{i-1}{n-1}$$
$$0.92 = \frac{i-1}{5-1}$$
$$0.92\times 4 = i-1$$
$$3.68 = i-1$$
$$i = 3.68+1$$
$$i = 4.68$$

$$\pi = 4.68\times 0.821\times 300$$
$$\pi = 5.76342 atm$$

16. (a) Write any two factor which affect conductance and conductive an electrolyte.

Sol:

(i) Nature of Electrolyte

(ii) Temperature

(iii) Concentration of solution

(b) Corrosion is electrochemical phenomena explain.

Sol:

Corrosion is electrochemical phenomena because this process Fe is oxide Fe^{+2}

Any work as anode part of iron where Fe oxidize into Fe^{+2} it work as anode and the electrode release that anode are accepted gain by H^+ ion to reduce into H^{2+} an makes cathode.

17. Given the M=0.10 KCl solution conductivity is
K=0.0129 s cm^{-1}
M=?

Sol.

$$M = \frac{K \times 1000}{M}$$

$$= \frac{0.129 \times 1000 \times 100}{0.10 \times 10000}$$

$$= 129 s \text{ cm}^2 mol^{-1}$$

18. 20% dissociation it takes minutes then how calculate the half-life.

Sol.

Let A- product

$t_{1/2}$ =?

$$n = \frac{a \times 20}{100} = \frac{2a}{10} = 0.2a$$

$$k = \frac{2.303}{t} \log \frac{a}{(a-x)}$$

$$k = \frac{2.303}{40} \log \frac{a}{(a-0.2a)}$$

$$= \log \frac{10}{8}$$

$$= \log 10 - \log 8$$

$$= 1 - \log 2^3$$

$$= 1 - 3 \log 2$$

$$= 1 - 3 \times 0.3010$$

$$= 0.97$$

$$k = \frac{2.303}{40} \times 0.97$$

$$k_{1/2} = \frac{0.693}{k}$$

$$= \frac{2.303 \times 0.3010 \times 40}{2.303 \times 0.97} = 124.084 \min ute$$

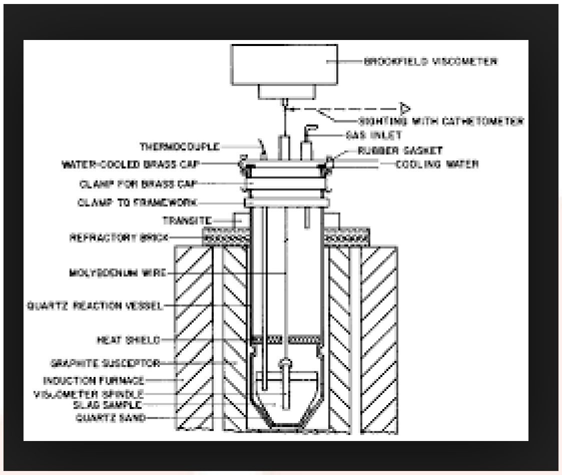
19. (a).What is the role of graphite rode in electrometallurgy of aluminium.

Sol.

In the electro metallurgy of aluminium graphite role work as anode.

(b) Draw the rubber batteries furnace figure.

Sol.



20. (i) $[CO(CN)_3]^{+3}$ **. Find the oxidation number of metal and coordination number.**

Sol.

$$[CO(CN)_3]^{+3} = x + 0 = 3$$

Oxidation number of metal =+3

Coordination number of metal =6

(ii) $K_4[Fe(CN)_6] = +4 + x + -6 = 0$

$x = +2$

Oxidation number of metal =+2

Coordination number of metal =6

20. Why phenol more acidic than alcohol.

Sol.

Phenoxide ion established and alcoxide ion is not established resonation. Hence stability of alcohol ion is less than that of a alcohol while this stability phoenix is more than phenol because phenol is more than acetic alcohol.

21. Esterification this equation.

(i) $CH_3 - CH_2 - OH$

(ii) $(CH_3)_3 CH - O - H$ $\quad$ $\mathrm{CH_3} - \underset{\mathrm{OH}}{\underset{|}{OH}} - CH_3$

(iii) $\left(CH_3\right)_3 C-O-H$ $\quad \begin{array}{c} \quad\quad\quad CH_3 \\ \quad\quad\quad | \\ CH_3 - \; C \; -OH \\ \quad\quad\quad | \\ \quad\quad\quad CH_3 \end{array}$

Sol.

$$\left(CH_3\right)_3 C-O-H < \left(CH_3\right)_3 - O-H < CH_3.CH_2-O-H$$

OR,

(i) **Why the boiling point of alcohol is much more than hydrocarbon, or ether having comparable molecular mass.**

Sol.

(ii) The boiling point of alcohol is much more than hydrocarbon, or ether having comparable molecular mass because the intermolecular hydrogen bonding is stronger than ether and hydro carbon**.**

(ii) Increasing order alcohol in dehydration below this equation.

$CH_3-CH_2-OH \quad (1^0)$

$\left(CH_3\right)_2 CH-O-H(2^0)$

$(CH_3)_3-C-O-H\left(3^0\right)$

Sol.

$$CH_3-CH_2-O-H < (CH_3)_2CH-O-H < (CH_3)_3-C-O-H$$

22. $C_6H_5-N_2^+Cl^- \xrightarrow[NaNO_2]{Cn} (A) \xrightarrow[6[H]]{Sn+HCl} (B)$

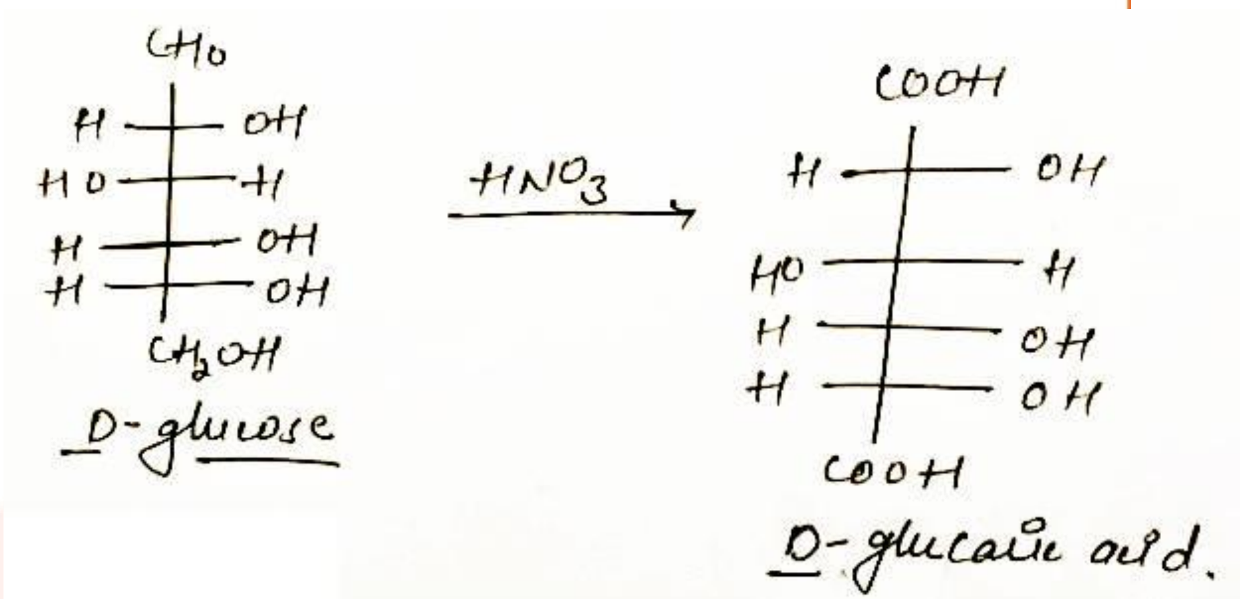
Sol:

$C_6H_5-N_2^+Cl^- \xrightarrow[NaNO_2]{Cn} C_6H_5-NO_2 \xrightarrow[6[H]]{Sn+HCl} C_6H_5-NH_2$

23.

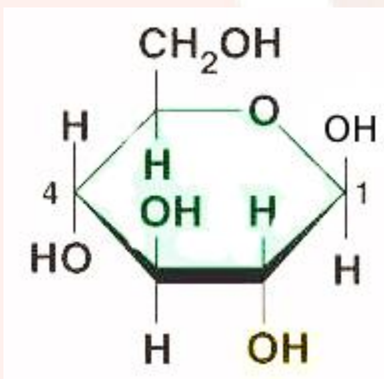
(a) What happens when, Glucose is reacted with conc HNO_3

Sol:



(b) **Write the structure of** $\beta - d$ ribose sugar

Sol:



β Glucose

24. (i) Differentiate between in configrational and conformational isomers.

Sol.

Conformational isomers are interconvert able at room temperature while consfigrational isomers cannot be inter convert able at room temperature.

(iii) Chair conformable and cyclohexene mostable both conformable .

Sol.

$$Chair \rightarrow C_1 - C_2, C_3 - C_4, C_5 - C_6 \left(stragered \text{ conformation} \right)$$

$$\mathrm{C}_{2-3},\ \mathrm{C}_{5-6} \text{ (Eclipsed conformation)}$$

Section C

25. (i) Why the basic nature of hydroxide for lanthanide decreases from moving left to right.

Sol.

The size of lanthanide due to lanthanide contration decrease lanthanide series on moving left to right as a result of this the nature of hydroxide of their decrease on moving left to right.

(ii) Any two lanthanide made by mismetal.

Sol.

(i) Ce (ii) Nd

(ii) Write the any lenthonide show a (+4) oxidation.

Sol.

Ce (cerium)

26. **(a) Why chemical are added in food.**

Sol.

Chemical substance can play an important role in food production and preservation. Prolong the shelf life of food, chemical used to fight diseases in farm animal or crops.

(b) Write the name of any two preservative food.

Sol.

(i) Sodium benzoate $\left(C_6H_3COONa\right)$

(ii) Propanoic acid

(c) Explain, the diabetic patient advised for using sekkerenee.

Sol.

The diabetic patient advised for using sekkerenee because sekkerenee enters directly in digestive system without providing food energy.

27.(a) write the structure formula oxalic acid .

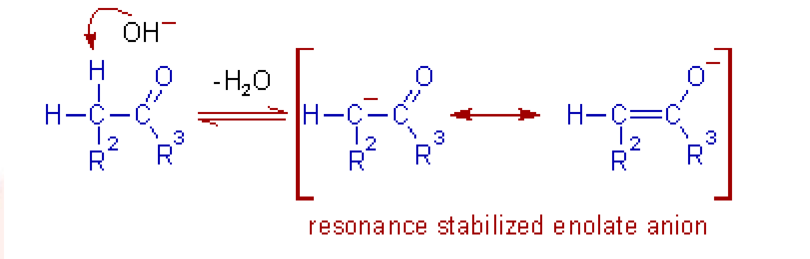
Sol.

$$\begin{array}{c} COOH \\ | \\ COOH \end{array}$$

(c) write the mechanism of aldol condensation

Sol:

In aldol condensation carbon-carbon bond is bond when enol molecules reacts with aldehyde or ketone.



OR,

(i) write the structure of diethyl ketone.

Sol.

$$C_2H_5-\underset{\|}{C}-C_2H_5$$

$$CH_3 \quad CH_2-\underset{\underset{O}{\|}}{C}-CH_2-CH_3$$

(ii) Explain the Kolbe electrolysis mechanism.

Sol.

Kolbe electrolysis also called as Kolbe reaction is an orgainic reaction . This reaction is an electrochemical decarboxylation of the salt of carboxylic acid which proudces radicals this reaction is used is synthesis of symmetrical dimers.

Section D

28. (a) what is definition of adsorption.

Sol.

Adsorption is a phenomena in which absorbent attract adsorbit molecule towards a surface an repine these molecules on its surface phenomena as known as adsorption.

(b) What happen when electric current is pass collide solution?

Sol.

When electric current is pass collide solution then it is coagulate.

(c) Why potash alm is used is purification of water.

Sol.

Al^{3+} obtain from potash alm neutralism the negatively charged in particle present in the water such that they are receipt and coagulated as a result water are purifiers.

(**c)**

OR,

(a) Define chemical adsorption.

Sol.

The absorption is which adsorbed molecules are bounded on the surface of the adsorvant by true chemical bond or by co.valent bond is known as chemical adsorption.

(b) What happen when light beam passes through colloidal solution.

Sol.

When light beam passes through colloidal solution then colloidal particle scattered the light which is known as Tindal effect as a result of this beam is clearly seen to the colloidal solution.

(c) why does the colour of sky appear in blue.

Sol.

Because light is scattered in all directions by the tiny molecules of air Earth's

The sky appear blue due to the scattering of light because the wavelength of blue light is very law.

29.(a) Write the Oxidation number of nitrogen in nitric acid HNO_3

Sol.

$$HNO_3 \rightarrow +1 + x + 3\times(-2) = 0$$

$$1+x-6=0$$

$$x=+5$$

(b) What happen when sulpher reacts with sulphuric acid H_2SO_4

Sol.

$$S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$$

(d) Why helium is used at place oxygen in modern equipment Collins diverse.

Sol.

Helium is used scuva tank because it is chemically inert that is to say it does not a participate a chemical reaction. Under water a scuba diver is subjected to added pressure. In shallow water the pressure difference is not so great, and one can simplify uses compressed air to breath.

(d) Write the Structure of $HClO_3$

Sol.

V.E = $1+7+6\times3=26$

$$= \frac{26}{8} = 3 + \frac{2}{2}$$
$$\Rightarrow 3+1=4$$
$$= sp^3$$

OR,

(a) What is the hybridization of nitrogen in NH_3**.**

Sol.

$NH_3 - sp^3$

(b) Write the equation of carbon form in H_2SO_4**.**

Sol.

$$C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$$

(c) Interhelogen compound more reaction than halogen.

Sol.

The inter halogens are generally more reactive than halogens except F. This is because A-X bonds in interhalogens are than the X-X bonds in dihalogen molecules. Reaction of inter halogens are similar to halogens. Hydrolysis of interhalogen compound give halogen acid and OXY-acid.

(d). Write the structure of XeF_6**.**

Sol.

V.E = $8+7\times 6=50$

$$\Rightarrow \frac{50}{8}=6+\frac{2}{2}$$
$$\Rightarrow 6+1=7$$
$$\Rightarrow sp^2d^3$$

30. (a) Write the chemical equation pinnacle state reaction.

Sol.

$$CH_3-Cr+NaI \xrightarrow{Acatone} CH_3-I-NaCr$$

(b) Why aerilhalide are less reactive towards nueophilics substitution reaction.

Sol.

(e) Ascending order from in SN.

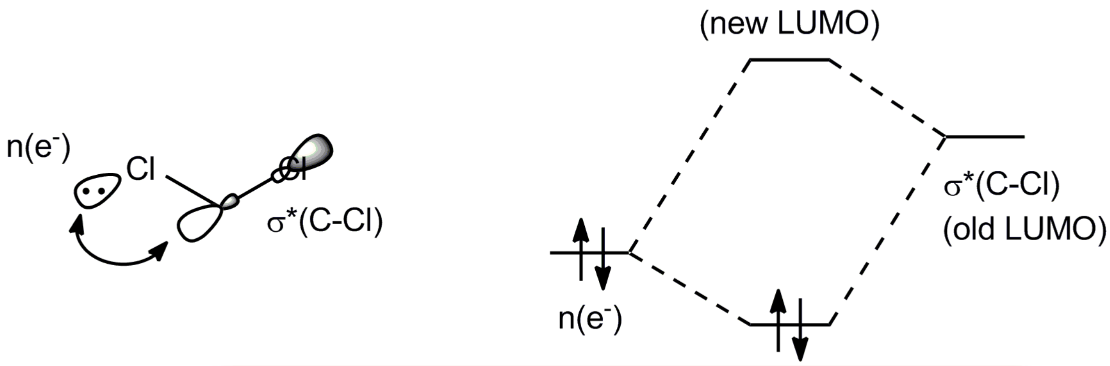
$$CH_3-CH_2-Cr,\left(CH_3\right)_2CH-Cr,(CH_3)_3-C-Cr$$

Sol.

$$1^0>2^0>3^0$$
$$(CH_3)_3-C-Cr<\left(CH_3\right)_2CH-Cr<CH_3-CH_2-Cr$$

(d). Draw the orbital diagram of methyl chloride.

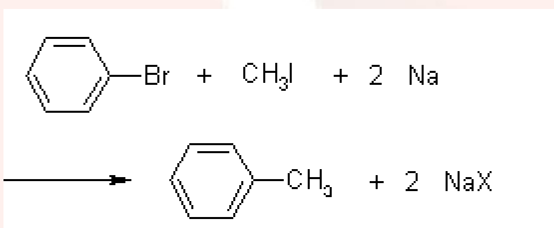
Sol.



OR,

(a) Write wurtz fitting reaction.

Sol.



(b) What happens when alkilehalide reacts with dilute KOH.

Sol.

$$CH_3 - CH_2 - Cl + KOH \rightarrow CH_3 - CH_2 - O - H$$
$$CH_3 - CH_2 - Cl + KOH \rightarrow CH_2 = CH_2$$

(c) Draw the labour diagram preparation of chloroform laboratory of chloroform.

Sol.

When chloroform is prepared by heating choral hydrate with strong solution of caustic soda. The mixture of bleaching power with ethanol after heating laboratory is prepared is chloroform. Chlorine gas evolved from bleaching powder paste oxides and chlorinates ethanol.

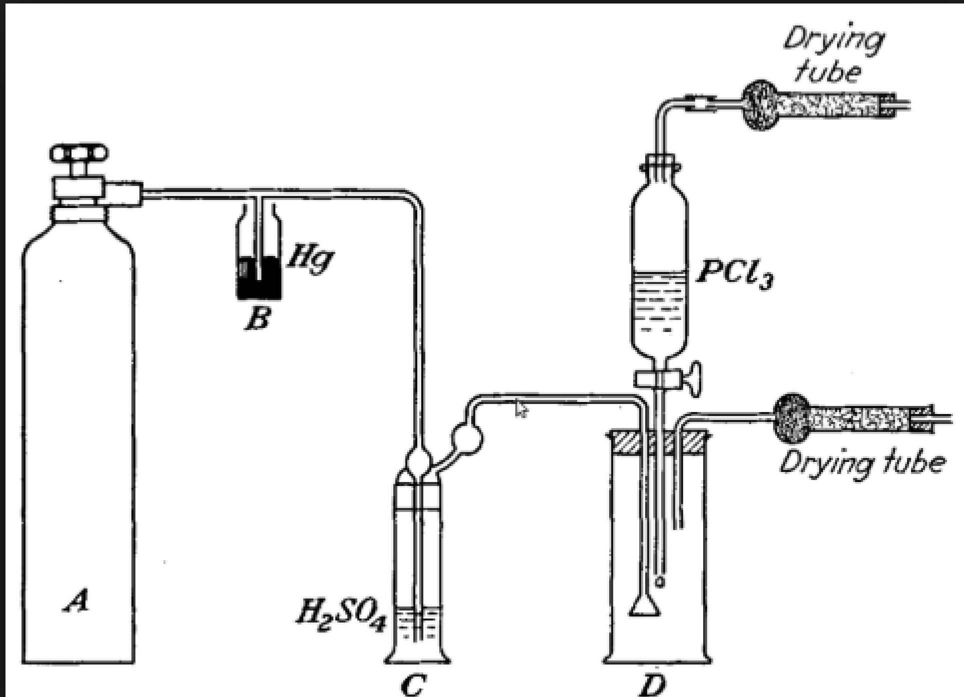


FIG. 16.—Preparation of phosphorus pentachloride.